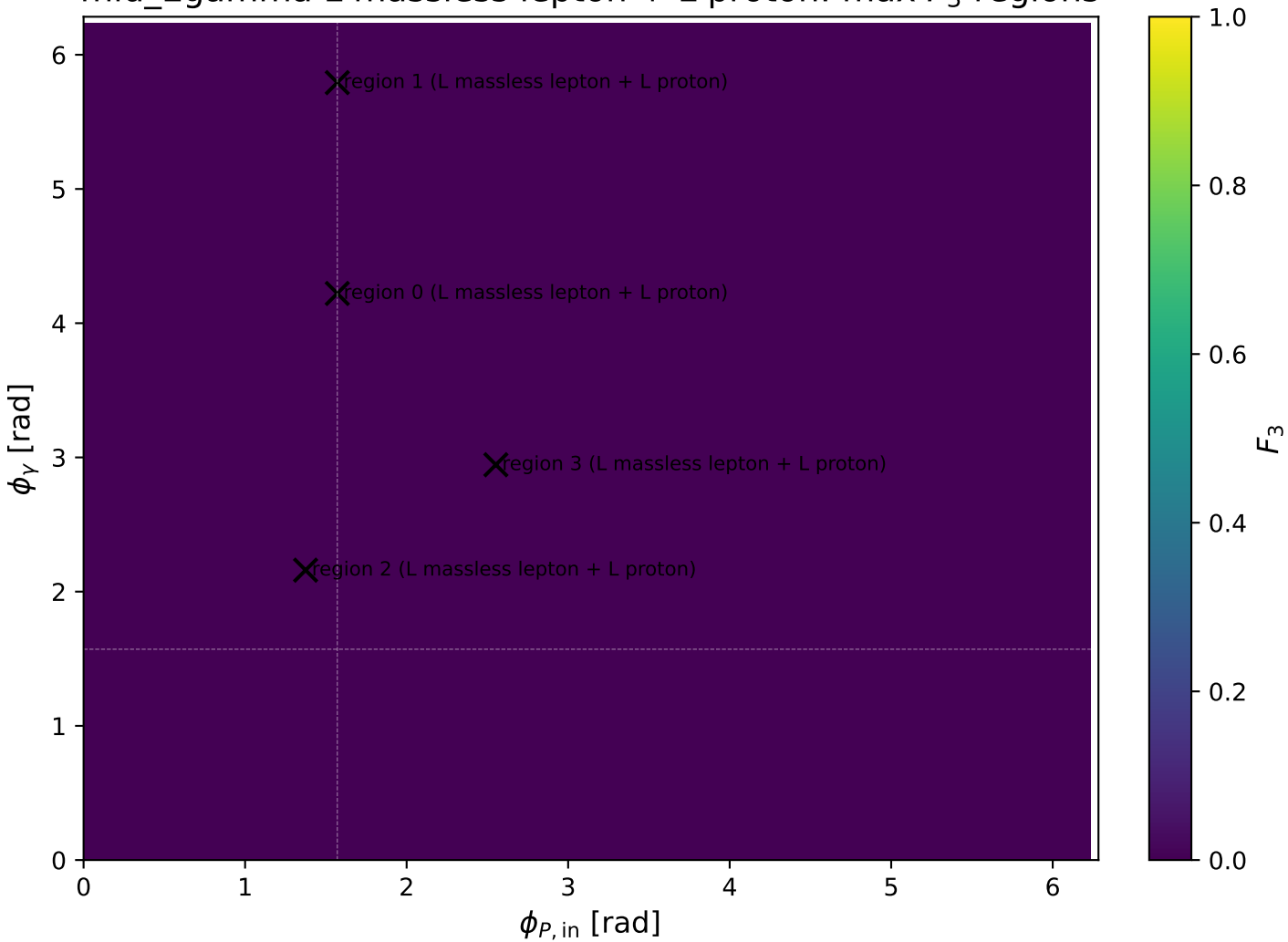
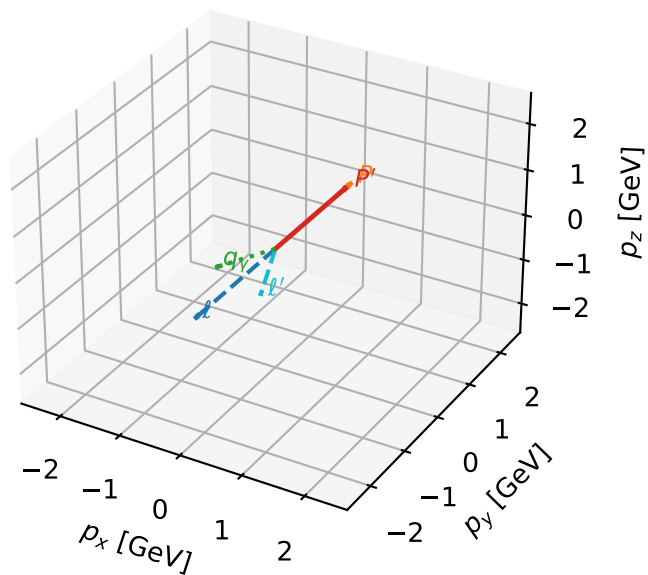


mid_Egamma L massless lepton + L proton: max F_3 regions



L massless lepton + L proton characteristic max F_3 configuration

3D momenta

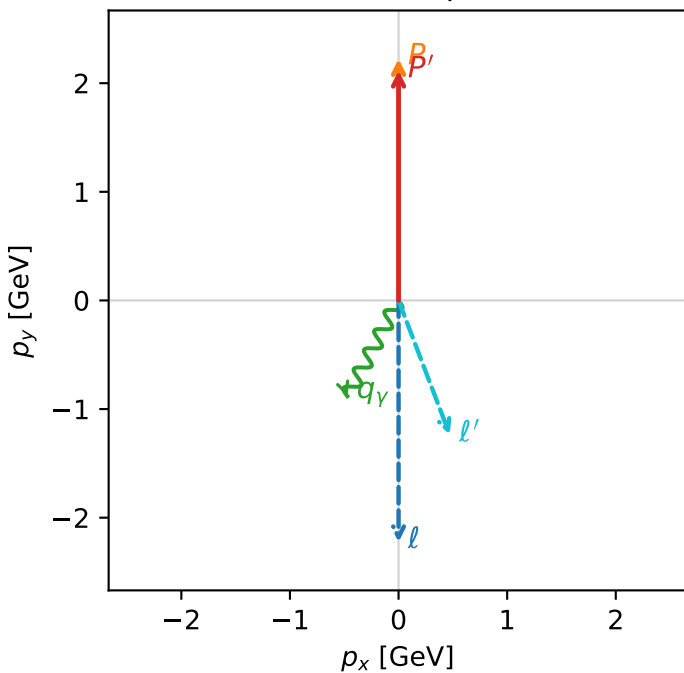


f3_mid_egamma_ll_region_0 F_3=1.50781e-08
 region: mid_Egamma_LL_region_0
 outgoing-state purity: 1
 kinematic point: high_s high_theta_in_mid_Egamma
 incoming state: $|L_{\ell_0}\rangle \otimes |L_p\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=2.11756$
 $\theta_{in}=1.57079$, $\phi_{\ell}=4.71239$, $\phi_p=1.5708$
 $E_Y=1$, $\phi_Y=4.22152$
 $Q^2=0.386451$, $x_B=0.044148$, $t=-0.00179699$, $W^2=9.24691$, $y=0.424187$
 $\theta(\ell', \gamma)=49.0069$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= -0.0000$ $p_y= -2.2244$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= 0.0000$ $p_y= 2.2244$ $p_z= 0.0000$
 ℓ' $E= 1.3225$ $p_x= 0.4714$ $p_y= -1.2356$ $p_z= 0.0000$
 P' $E= 2.3160$ $p_x= 0.0000$ $p_y= 2.1176$ $p_z= 0.0000$
 q_Y $E= 1.0000$ $p_x= -0.4714$ $p_y= -0.8819$ $p_z= 0.0000$

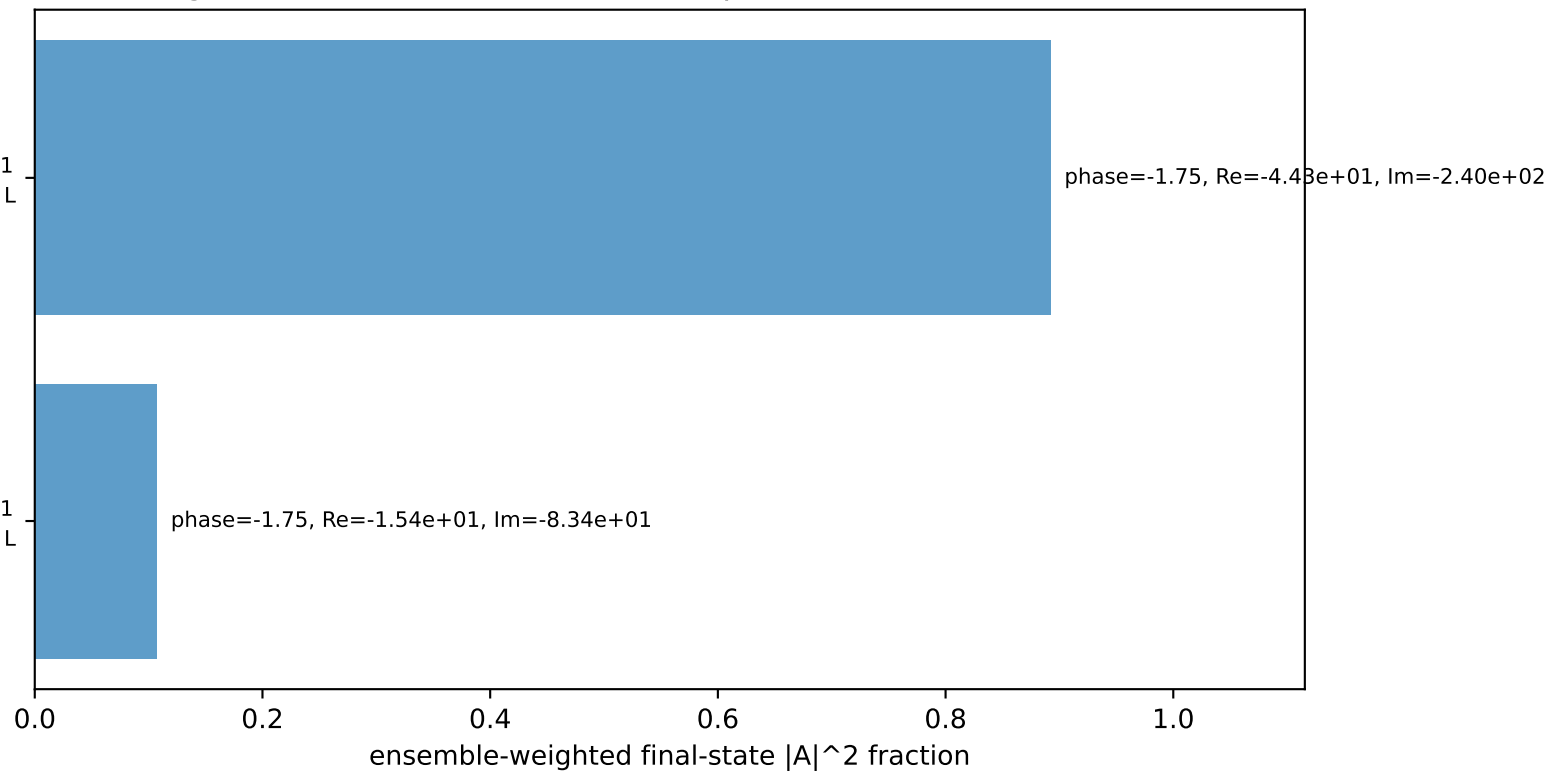
Transverse plane



$h_e=+1, h_p=-1, h_Y=+1$
 massless lepton L, proton L

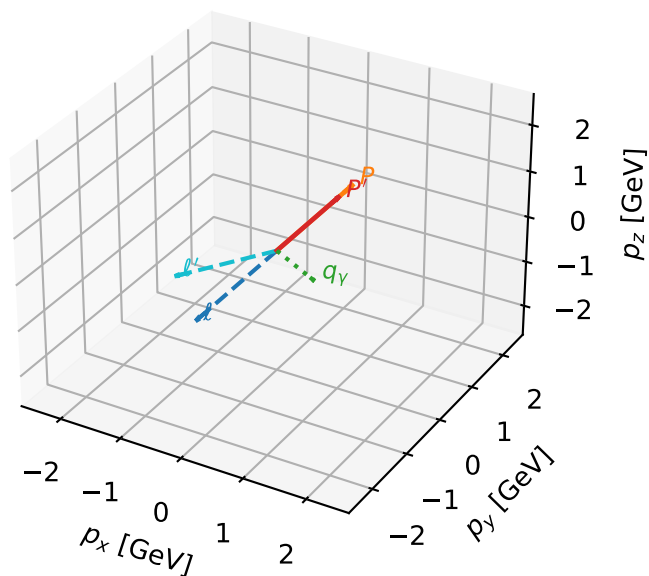
$h_e=+1, h_p=+1, h_Y=-1$
 massless lepton L, proton L

Leading initial-ensemble/final-state components (N=2, retained=100.0%)



L massless lepton + L proton characteristic max F_3 configuration

3D momenta

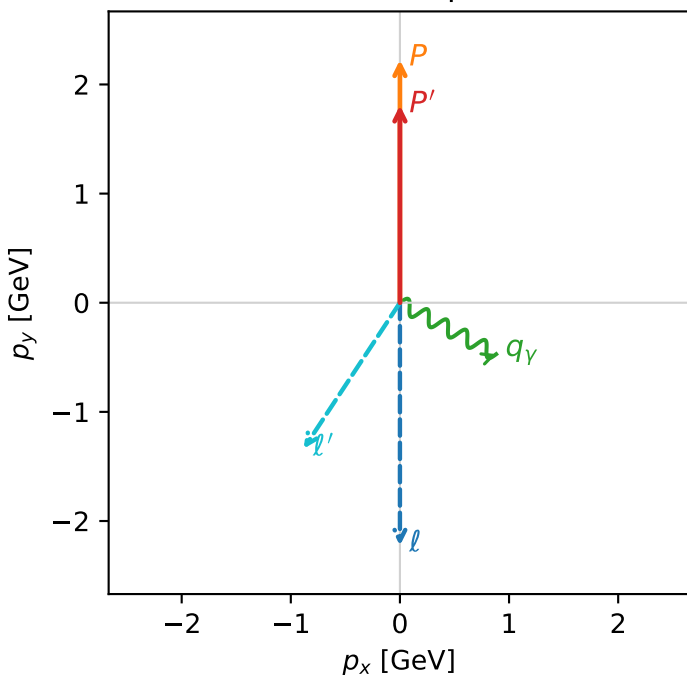


f3_mid_egamma_ll_region_1 F_3=7.91753e-09
region: mid_Egamma_LL_region_1
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $|L_{\ell_0}\rangle \otimes |L_p\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=1.8082$
 $\theta_{in}=1.57079$, $\phi_{\ell}=4.71239$, $\phi_P=1.5708$
 $E_Y=1$, $\phi_Y=5.79231$
 $Q^2=1.17763$, $x_B=0.169287$, $t=-0.0310449$, $W^2=6.65862$, $y=0.337101$
 $\theta(\ell', \gamma)=95.2889$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= -0.0000$ $p_y= -2.2244$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= 0.0000$ $p_y= 2.2244$ $p_z= 0.0000$
 ℓ' $E= 1.6015$ $p_x= -0.8819$ $p_y= -1.3368$ $p_z= 0.0000$
 P' $E= 2.0370$ $p_x= 0.0000$ $p_y= 1.8082$ $p_z= 0.0000$
 q_Y $E= 1.0000$ $p_x= 0.8819$ $p_y= -0.4714$ $p_z= 0.0000$

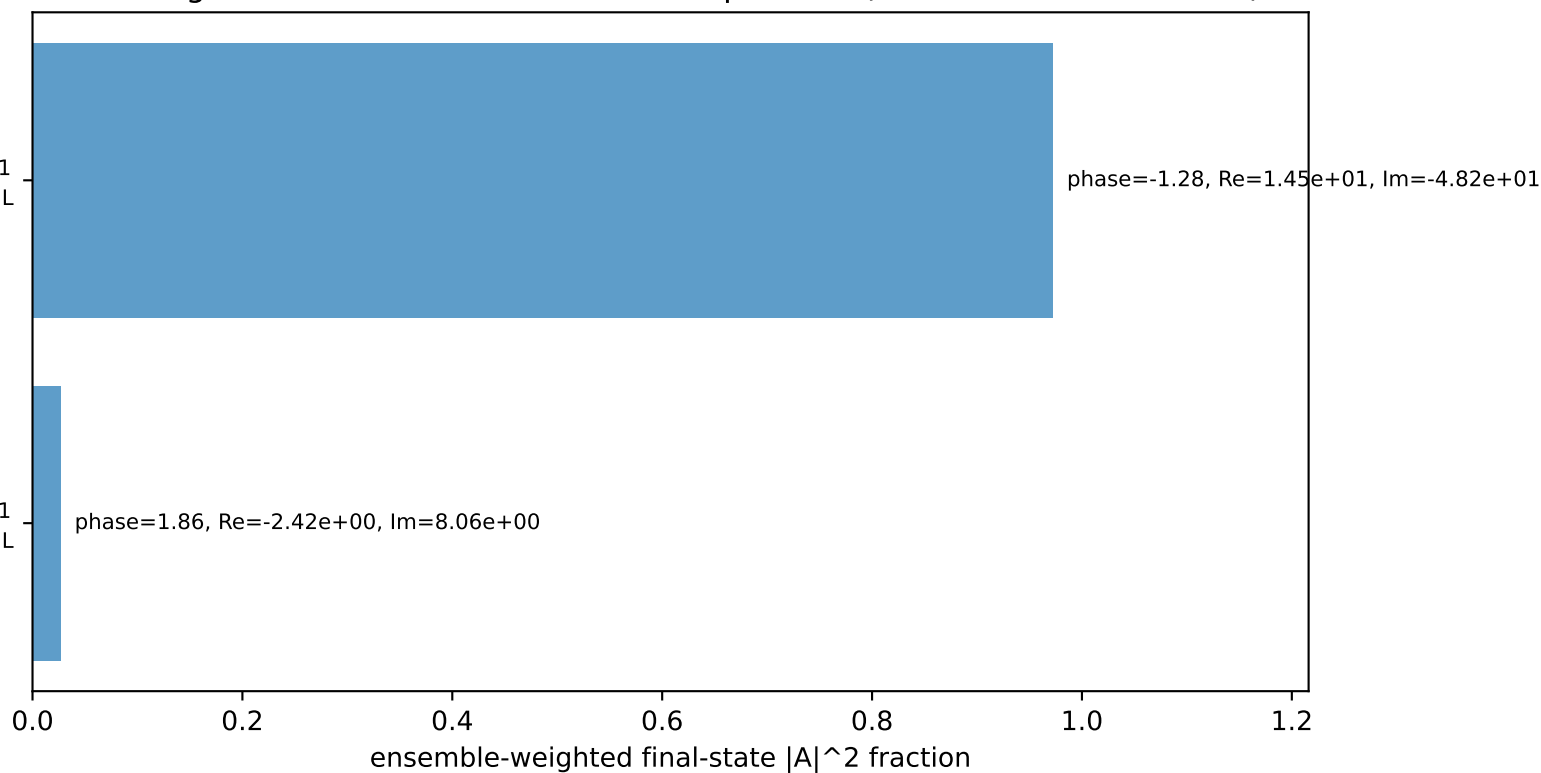
Transverse plane



$h_e=+1, h_p=-1, h_Y=+1$
massless lepton L, proton L

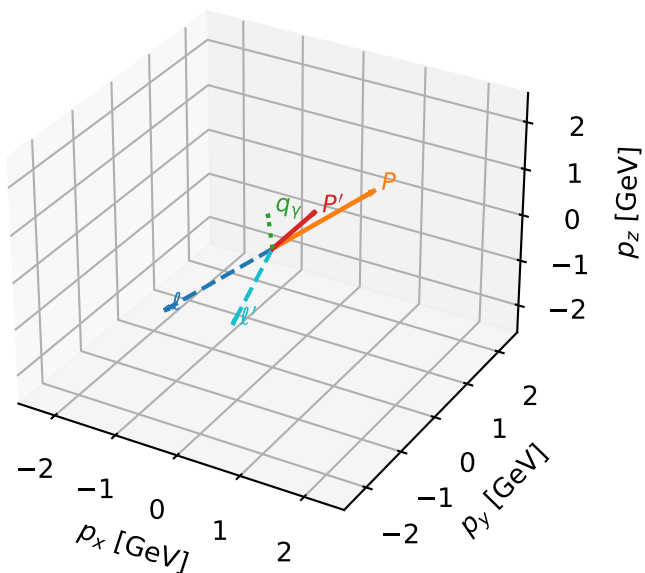
$h_e=+1, h_p=+1, h_Y=-1$
massless lepton L, proton L

Leading initial-ensemble/final-state components (N=2, retained=100.0%)



L massless lepton + L proton characteristic max F_3 configuration

3D momenta

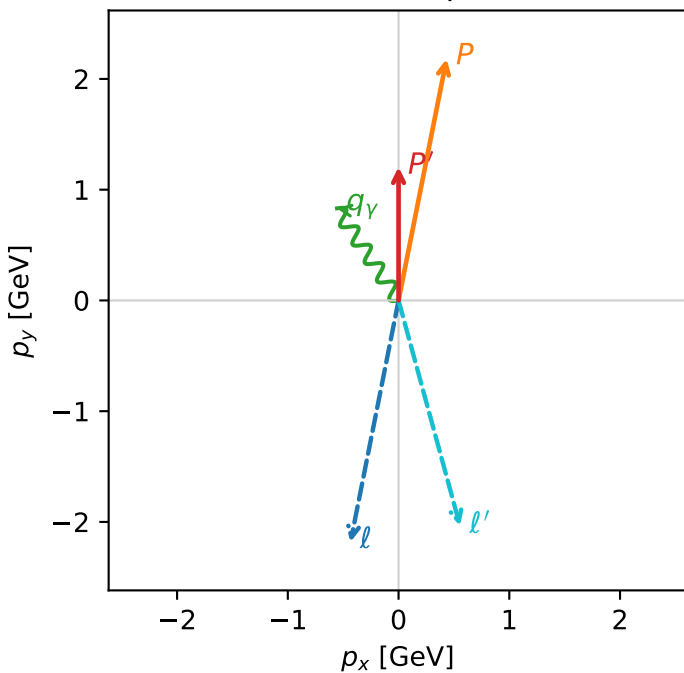


f3_mid_egamma_ll_region_2 $F_3=6.19169\text{e-}09$
 region: mid_Egamma_LL_region_2
 outgoing-state purity: 1
 kinematic point: high_s high_theta_in_mid_Egamma
 incoming state: $|L_{\ell_0}\rangle \otimes |L_p\rangle$

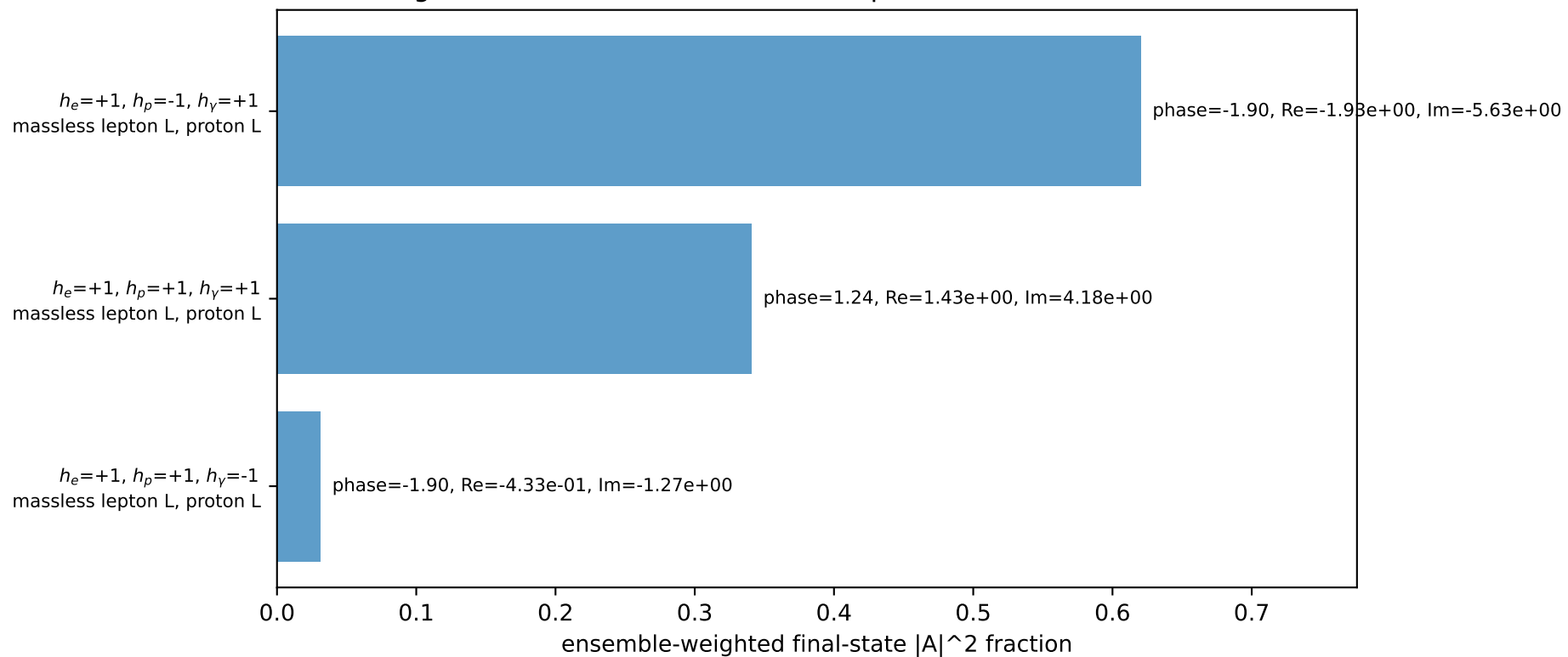
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=1.20533$
 $\theta_{\text{in}}=1.57079$, $\phi_\ell=4.51604$, $\phi_P=1.37445$
 $E_\gamma=1$, $\phi_\gamma=2.15984$
 $Q^2=0.987348$, $x_B=0.484571$, $t=-0.355172$, $W^2=1.93007$, $y=0.0987386$
 $\theta(\ell', \gamma)=161.507$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= -0.4340$ $p_y= -2.1817$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= 0.4340$ $p_y= 2.1817$ $p_z= 0.0000$
 ℓ' $E= 2.1112$ $p_x= 0.5556$ $p_y= -2.0368$ $p_z= 0.0000$
 P' $E= 1.5273$ $p_x= 0.0000$ $p_y= 1.2053$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= -0.5556$ $p_y= 0.8315$ $p_z= 0.0000$

Transverse plane

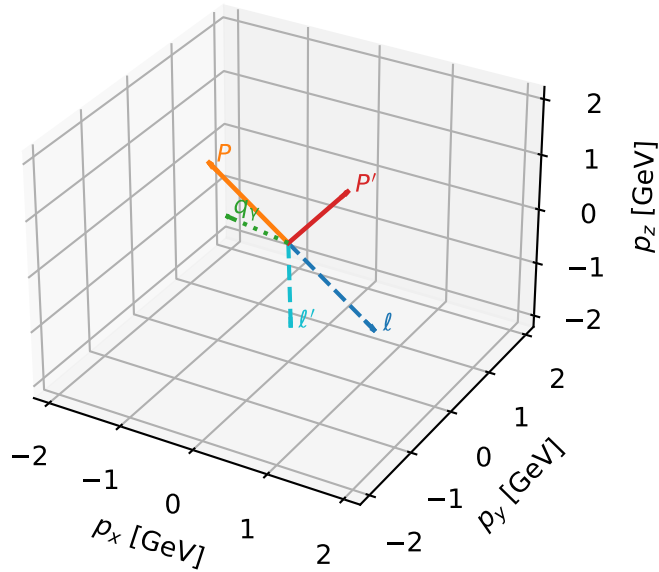


Leading initial-ensemble/final-state components (N=3, retained=99.3%)



L massless lepton + L proton characteristic max F_3 configuration

3D momenta

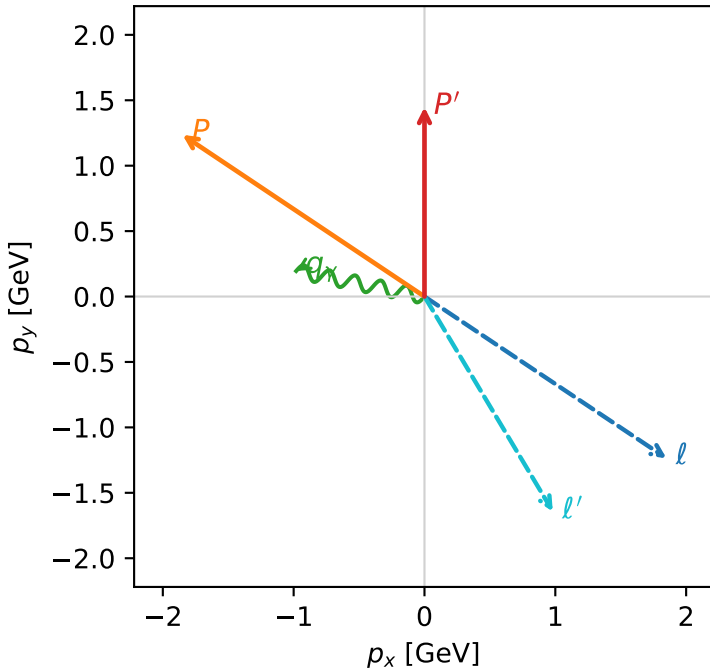


f3_mid_egamma_ll_region_3 $F_3=3.65597\text{e-}09$
 region: mid_Egamma_LL_region_3
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|L_{l_0}\rangle \otimes |L_p\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=1.44782$
 $\theta_{in}=1.57079$, $\phi_l=5.69414$, $\phi_p=2.55254$
 $E_\gamma=1$, $\phi_\gamma=2.94524$
 $Q^2=0.823719$, $x_B=0.222084$, $t=-2.99103$, $W^2=3.76517$, $y=0.179736$
 $\theta(l', \gamma)=132.086$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2244$ $p_x= 1.8495$ $p_y= -1.2358$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= -1.8495$ $p_y= 1.2358$ $p_z= 0.0000$
 l' $E= 1.9134$ $p_x= 0.9808$ $p_y= -1.6429$ $p_z= 0.0000$
 P' $E= 1.7251$ $p_x= 0.0000$ $p_y= 1.4478$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= -0.9808$ $p_y= 0.1951$ $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=4, retained=100.0%)

